

Course Activities - Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Each activity has its own self-contained folder with a procedure, scenario, worksheet, evidence checklist and same-folder printable PDFs.

1. [Device Physics Evidence Map](#) - K1 | A1 - 35 minutes
2. [Process Module Architecture](#) - A1 - 40 minutes
3. [CMOS Manufacturing Flow and Control Gates](#) - A2 - 50 minutes
4. [Lithography-Etch Interaction Case](#) - A3 - 45 minutes
5. [Doping and Thermal-Budget Interaction](#) - A3 - 45 minutes
6. [Yield and Performance Metric Verification](#) - K2 | A4 - 55 minutes
7. [Reliability Failure Analysis](#) - K2 | A4 - 45 minutes
8. [Follow-Up Action and Integration Review](#) - A5 - 60 minutes

Assessment alignment

- Activities 1-3 prepare evidence for Case Study Task 1 (A1-A2).
- Activities 4-8 prepare evidence for Case Study Task 2 (A3-A5).
- Activities 6-7 reinforce K2 performance metrics used in the Written Assessment.